to structure your authentication flow, and the best approach depends on the specific requirements of your application. Detailed instructions for configuring Clerk using this approach are located in the appendix.

I've established these policies to mirror real-world use cases and provide a baseline for securing your AI web application. However, feel free to adjust the policy limits, the message quota, allowed authentication methods, or specific route protections. Consider factors such as the cost of your AI resources or the anticipated usage patterns of your users when making these adjustments.

Project code: chat-authentication

The code for this example project is in the ch09/chat-authentication folder. You can use that example by running this command in the root folder of the project repository:

```
$ npm run dev -w ch09/chat-authentication
```

You'll need the Google API key, the existing Upstash keys and the new Clerk.js authentication keys to access their respective REST APIs. These are configured as environment variables:

```
// .env
GEMINI_API_KEY=<GEMINI_AI_SECRET_KEY>
UPSTASH_REDIS_REST_URL=<UPSTASH_REDIS_REST_URL>
UPSTASH_REDIS_REST_TOKEN=<UPSTASH_REDIS_REST_TOKEN>
NEXT_PUBLIC_CLERK_PUBLISHABLE_KEY=<NEXT_PUBLIC_CLERK_PUBLISHABLE_KEY>
CLERK_SECRET_KEY=<CLERK_SECRET_KEY>
```

Instructions on how to set up and create secret keys and necessary credentials for Google AI and Upstash Redis, as well as how to retrieve your Clerk API keys, are provided in the appendix.

CLERK PROJECT SETUP

Create a new Clerk project at the Clerk.js website. Configure the desired authentication methods (e.g., username/password). At the end of setup, you must populate the following .env variables for Clerk to work:

```
CLERK_SECRET_KEY=
NEXT_PUBLIC_CLERK_PUBLISHABLE_KEY=
```

MIDDLEWARE CONFIGURATION

Next, we need to configure the authentication middleware in Next.js. Install the necessary Clerk packages in your Next.js project `@clerk/nextjs` using npm or Yarn. Then create a middleware.js file in the root of your project (or in the `/src` directory if you're using one).